Numerical Methods for Solving Nonlinear Equations and Systems of Nonlinear Equations

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Preface

This thesis has been prepared as part of the undergraduate studies in Mathematical Engineering at Yıldız Technical University.

The main motivation behind this work is the author's interest in numerical methods and optimization techniques. During the study, an effort has been made to understand the fundamental principles of solving nonlinear equations and to explore their practical implementations.

The author would like to express sincere gratitude to the advisor, Prof. Dr. Hale Gonce Köçken, for her guidance, support, and valuable insights throughout the preparation of this thesis.

Abstract

What is Done During This Term

During this term, a comprehensive review of the literature related to numerical methods for solving nonlinear equations has been conducted. Fundamental concepts such as root-finding, fixed-point iteration, convergence, and Jacobian-based methods have been studied to establish a solid theoretical background. The importance, scope, and objectives of the thesis have been clearly defined, focusing on both single nonlinear equations and systems of nonlinear equations.

In addition, a set of numerical methods to be analyzed in the thesis has been determined, including bisection, secant, Newton, damped Newton, and Brent methods. The general structure of the thesis has been established in accordance with academic writing standards, and the initial chapters covering the introduction and mathematical background have been completed.

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1. Introduction

1.1 Motivation Behind Numerical Methods

When attempting to solve a mathematical problem, two general approaches can be considered: analytical solutions and numerical solutions. Analytical solution methods aim to obtain the solution of a given problem exactly and in closed form. Many classical mathematical problems can be solved using such methods, allowing the exact value of the solution to be determined.

However, for a significant portion of problems encountered in practice, analytical solutions either do not exist or are too costly to obtain. In particular, when dealing with nonlinear equations and systems of nonlinear equations, closed-form solutions are often not available. In such cases, numerical methods are employed.

Rather than obtaining the exact solution directly, numerical methods aim to compute an approximate value of the solution with a certain level of accuracy. These methods typically rely on iterative algorithms that progressively improve the approximation step by step until the desired level of accuracy is achieved.

Different numerical methods have their own advantages and disadvantages in terms of properties such as convergence rate, computational cost, and stability. Therefore, selecting an appropriate method for a given problem plays an important role in the study of numerical analysis [1, 2].

1.2 Nonlinear Equations and Systems

A nonlinear equation is an equation of the form

$$f(x) = 0, \tag{1.1}$$

where f is a nonlinear function. In contrast to linear equations, nonlinear equations may exhibit multiple roots, no real roots, or complex solution structures, making their analysis and solution more challenging [1, 2].

A simple example of a nonlinear equation is

$$f(x) = x^3 - x - 2 = 0 \tag{1.2}$$

for which no closed-form analytical solution exists in a simple form. Such equations commonly arise in various scientific and engineering applications, including physics, optimization, and computational modeling.

In many practical cases, it is not possible to determine the exact solution of a nonlinear equation analytically. Even when an analytical solution exists, it may be too

complicated or computationally expensive to evaluate. For this reason, numerical methods are employed to approximate the roots of nonlinear equations with a desired level of accuracy.

The problem can be extended to systems of nonlinear equations. A nonlinear system can be written in vector form as

$$F(\mathbf{x}) = \mathbf{0}, \tag{1.3}$$

where

$$F : \mathbb{R}^n \to \mathbb{R}^n \tag{1.4}$$

is a vector-valued nonlinear function.

For example, consider the system

$$\begin{cases} x^2 + y^2 - 4 = 0, \\ x - y - 1 = 0. \end{cases} \tag{1.5}$$

Such systems arise naturally in applications where multiple variables interact in a nonlinear manner.

Solving nonlinear systems is generally more complex than solving single nonlinear equations. In particular, the solution methods often require the use of the Jacobian matrix, defined as

$$J_F(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}. \tag{1.6}$$

which generalizes the concept of the derivative to higher dimensions [3]. In this thesis, both single nonlinear equations and systems of nonlinear equations are considered. The focus is on numerical methods that iteratively approximate the solution and their behavior in terms of convergence, computational cost, and robustness.

1.3 Objectives of the Thesis

The main objective of this thesis is to study numerical methods for solving nonlinear equations and systems of nonlinear equations. These problems arise frequently in various fields of science and engineering, and in many cases, their exact solutions cannot be obtained analytically. As a result, numerical approaches play a fundamental role in approximating their solutions [1, 2].

In this context, the thesis focuses on two main problem classes. First, numerical methods for solving single nonlinear equations of the form $f(x) = 0$ are investigated. Classical methods such as bisection, secant, and Newton-type methods are examined in terms of their underlying principles and convergence behavior.

Second, the study extends to systems of nonlinear equations of the form $F(\mathbf{x}) = \mathbf{0}$. These systems are generally more complex and require more advanced techniques, particularly those involving the Jacobian matrix. Methods for solving such systems are analyzed within a similar theoretical framework.

In addition to the theoretical analysis, the selected numerical methods are implemented using a computational tool. The implementations are used to evaluate the practical performance of the methods through numerical experiments. In particular, the

methods are compared in terms of convergence behavior, computational efficiency, and robustness.

Overall, the aim of this thesis is to provide a clear and structured examination of numerical methods for nonlinear problems, combining theoretical foundations with computational analysis.

1.4 Structure of the Thesis

This section will provide an overview of the structure of the thesis, briefly describing the content of each chapter.

[This section will be completed after the thesis is finalized.]

2. Mathematical Background

2.1 Functions and Roots

In numerical analysis, the problem of solving a nonlinear equation is commonly expressed as finding a root of a function. A root of a function $f(x)$ is a value $x^* \in \mathbb{R}$ such that

$$f(x^*) = 0. \tag{2.1}$$

Depending on the structure of the function, there may exist a single root, multiple roots, or no real roots. In many practical applications, the function f is continuous, and root-finding methods often rely on this property [1, 2].

2.2 Fixed-Point Formulation

Many numerical methods are based on transforming the root-finding problem into a fixed-point problem. Instead of solving $f(x) = 0$, the equation is rewritten in the form

$$x = g(x), \tag{2.2}$$

where g is a suitably defined function.

A point x^* is called a fixed point of g if

$$x^* = g(x^*). \tag{2.3}$$

If g satisfies certain conditions, iterative methods of the form

$$x_{k+1} = g(x_k) \tag{2.4}$$

can be used to approximate the solution [1].

2.3 Convergence of Iterative Methods

An iterative method generates a sequence $\{x_k\}$ that is expected to approach the true solution x^*. The method is said to converge if

$$\lim_{k\to\infty} x_k = x^*. \tag{2.5}$$

The convergence of an iterative method depends on several factors, including the choice of the initial guess and the properties of the function.

A common condition for convergence in fixed-point iteration is that the function g is a contraction, meaning that there exists a constant $L < 1$ such that

$$|g(x) - g(y)| \leq L|x - y| \tag{2.6}$$

for all x, y in a neighborhood of the fixed point [1, 4].

2.4 Order of Convergence

The speed at which an iterative method converges is described by its order of convergence. Let $e_k = |x_k - x^*|$ denote the error at iteration k. The method is said to have order p if there exists a constant $C > 0$ such that

$$e_{k+1} \approx C e_k^p \tag{2.7}$$

for sufficiently large k.

For example:

- $p = 1$: linear convergence
- $p = 2$: quadratic convergence

Higher-order methods generally converge faster, but they may require more computational effort per iteration [1].

2.5 Norms and Vector Functions

For systems of nonlinear equations, it is necessary to measure the size of vectors. A commonly used norm is the Euclidean norm, defined as

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}. \tag{2.8}$$

Norms are used to quantify errors and convergence in higher-dimensional problems [2].

2.6 Jacobian Matrix

For a system of nonlinear equations $F(\mathbf{x}) = \mathbf{0}$, the Jacobian matrix plays a central role in many numerical methods.

The Jacobian matrix is defined as

$$J_F(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}. \tag{2.9}$$

It generalizes the concept of the derivative to multivariable functions and is essential in methods such as Newton's method for systems [3].

2.7 Stopping Criteria

In practical computations, iterative methods cannot run indefinitely. Therefore, a stopping criterion is required.

Common stopping criteria include:

- $|x_{k+1} - x_k| < \varepsilon$
- $|f(x_k)| < \varepsilon$
- maximum number of iterations reached

where ε is a predefined tolerance [1, 4].

3. Single Nonlinear Equations

3.1 Bisection Method

The bisection method is one of the simplest and most reliable numerical techniques for solving nonlinear equations of the form $f(x) = 0$. It is based on the Intermediate Value Theorem, which states that if a continuous function f changes sign over an interval $[a, b]$, then there exists at least one root in that interval [1, 2].

More precisely, if

$$f(a) \cdot f(b) < 0, \tag{3.1}$$

then there exists a point $c \in (a, b)$ such that

$$f(c) = 0. \tag{3.2}$$

The method starts with such an interval $[a, b]$ and repeatedly bisects it. At each iteration, the midpoint is computed as

$$c_k = \frac{a_k + b_k}{2}. \tag{3.3}$$

The value of the function at the midpoint, $f(c_k)$, is then evaluated. Depending on the sign of $f(c_k)$, the interval is updated as follows:

- If $f(a_k) \cdot f(c_k) < 0$, then the root lies in $[a_k, c_k]$, so set $b_{k+1} = c_k$.
- Otherwise, the root lies in $[c_k, b_k]$, so set $a_{k+1} = c_k$.

This process is repeated until a stopping criterion is satisfied, such as the interval length becoming sufficiently small or the function value approaching zero.

One of the main advantages of the bisection method is its guaranteed convergence. As long as the function is continuous and the initial interval satisfies the sign change condition, the method will always converge to a root [1].

However, the method has a relatively slow rate of convergence. The error decreases linearly, and the interval length is halved at each iteration. More precisely, after k iterations, the error satisfies

$$|x_k - x^*| \leq \frac{b_0 - a_0}{2^k}. \tag{3.4}$$

This linear convergence makes the method less efficient compared to methods such as Newton's method or the secant method. Nevertheless, due to its robustness and simplicity, the bisection method is often used as a baseline or as a component in more advanced hybrid methods such as Brent's method [2, 4].

3.2 Secant Method

The secant method is an iterative numerical technique used to solve nonlinear equations of the form $f(x) = 0$. It can be seen as a derivative-free alternative to Newton's method, as it approximates the derivative using finite differences instead of requiring an explicit analytical expression [1, 2].

Given two initial approximations x_0 and x_1, the method constructs a secant line through the points $(x_{k-1}, f(x_{k-1}))$ and $(x_k, f(x_k))$. The next approximation is obtained by finding the root of this secant line:

$$x_{k+1} = x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}. \tag{3.5}$$

This formula eliminates the need for computing derivatives, making the method useful when the derivative of the function is difficult or expensive to evaluate.

Unlike the bisection method, the secant method does not require the initial guesses to bracket a root. However, this also means that convergence is not guaranteed. The method may fail to converge if the initial guesses are not chosen appropriately or if the function behaves poorly in the neighborhood of the root [3].

In terms of convergence speed, the secant method is generally faster than the bisection method. It has a superlinear rate of convergence, with order approximately equal to

$$p \approx 1.618, \tag{3.6}$$

which is known as the golden ratio. Although this is slower than the quadratic convergence of Newton's method, it often provides a good balance between efficiency and computational cost [1].

One limitation of the secant method is that it requires two initial values and may encounter numerical instability if $f(x_k) \approx f(x_{k-1})$, which can lead to division by very small numbers. Despite these limitations, the method is widely used in practice due to its simplicity and effectiveness [4].

3.3 Newton's Method

Newton's method, also known as the Newton–Raphson method, is one of the most widely used techniques for solving nonlinear equations of the form $f(x) = 0$. Unlike the bisection and secant methods, Newton's method makes explicit use of the derivative of the function to achieve faster convergence [1, 2, 3].

Starting from an initial guess x_0, the method generates a sequence $\{x_k\}$ using the iterative formula

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}. \tag{3.7}$$

Geometrically, this update rule corresponds to approximating the function by its tangent line at x_k and taking the root of this linear approximation as the next iterate.

One of the main advantages of Newton's method is its rapid convergence. Under suitable conditions, such as when the initial guess is sufficiently close to the true root and $f'(x^*) \neq 0$, the method exhibits quadratic convergence. This means that the number of correct digits approximately doubles at each iteration [1].

However, the performance of Newton's method strongly depends on the choice of the initial guess. If the initial value is far from the root, the method may converge slowly, diverge, or converge to a different root. In addition, the method requires the computation of the derivative $f'(x)$, which may be difficult or computationally expensive for some functions [3].

Another limitation arises when the derivative is close to zero, which can lead to large updates and numerical instability. Despite these challenges, Newton's method remains a fundamental tool in numerical analysis due to its efficiency and strong theoretical properties [5].

3.4 Damped Newton Method

The damped Newton method is a modification of the classical Newton's method designed to improve its stability and convergence behavior. While the standard Newton method can converge very rapidly near the solution, it may fail or diverge if the initial guess is not sufficiently close to the root. The damped version addresses this issue by introducing a step size parameter [3, 5].

The iteration formula is given by

$$x_{k+1} = x_k - \alpha_k \frac{f(x_k)}{f'(x_k)}, \tag{3.8}$$

where $\alpha_k \in (0, 1]$ is a damping parameter. When $\alpha_k = 1$, the method reduces to the classical Newton's method.

The main idea is to control the size of the update step in order to ensure more stable convergence. In practice, α_k is often chosen using a line search strategy, which selects a step size that sufficiently reduces the function value or a related merit function [3].

One common approach is to require that the function value decreases at each iteration, that is,

$$|f(x_{k+1})| < |f(x_k)|. \tag{3.9}$$

This condition helps prevent large and potentially unstable updates, especially when the derivative is small or when the function is highly nonlinear.

Compared to the standard Newton method, the damped version generally converges more reliably from a wider range of initial guesses. However, this increased robustness comes at the cost of slower convergence, particularly when the method is already close to the solution.

The damped Newton method provides a useful balance between efficiency and stability, and it also establishes a direct connection with optimization techniques, where similar step size control strategies are widely used.

3.5 Brent's Method

Brent's method is a hybrid root-finding algorithm that combines the reliability of bracketing methods, such as the bisection method, with the efficiency of interpolation-based methods like the secant method. It is widely regarded as one of the most effective and robust methods for solving nonlinear equations in practice [2, 5].

The method maintains an interval $[a, b]$ such that $f(a) \cdot f(b) < 0$, ensuring that a root is always bracketed. At each iteration, Brent's method attempts to use a faster interpolation technique, such as the secant method or inverse quadratic interpolation, to estimate the root. If the interpolation step is not reliable, the method falls back to the bisection step to guarantee convergence [4].

Unlike pure methods, Brent's method dynamically selects between different strategies based on the behavior of the function. This adaptive approach allows it to achieve both robustness and efficiency.

In inverse quadratic interpolation, three points are used to construct a quadratic approximation of the inverse function, and the root of this approximation is taken as the next iterate. This typically leads to faster convergence compared to linear interpolation methods such as the secant method.

One of the key advantages of Brent's method is that it does not require the computation of derivatives while still achieving a convergence rate that is often comparable to Newton-type methods. At the same time, the bracketing condition ensures that the method will not diverge, making it more reliable than open methods such as the secant or Newton methods [5].

Due to this combination of speed and robustness, Brent's method is commonly used in practical numerical libraries and is often preferred when both reliability and efficiency are required.

3.6 Illustrative Example

In order to make the behavior of the methods more concrete, the same nonlinear equation is used throughout this chapter:

$$f(x) = x^3 - x - 2 = 0. \tag{3.10}$$

Since

$$f(1) = -2 \qquad \text{and} \qquad f(2) = 4,$$

there is at least one real root in the interval $[1, 2]$. The approximate root is

$$x^* \approx 1.52138.$$

This example is suitable for all five methods discussed in this chapter. The aim here is not to give a fully detailed manual computation for every iteration, but to show a short sequence of iterates so that the convergence behavior can be observed clearly.

3.6.1 Bisection Method

For the bisection method, the initial interval is taken as $[1, 2]$. At each step, the midpoint of the current interval is computed and the interval is updated according to the sign of the function.

The method converges steadily, and the root always remains inside the updated interval. However, the progress is relatively slow, since the interval length is only halved at each step.

Table 3.1: Bisection method for $f(x) = x^3 - x - 2$

Iteration	a_k	b_k	Midpoint c_k	$f(c_k)$
1	1.00000	2.00000	1.50000	-0.12500
2	1.50000	2.00000	1.75000	1.60938
3	1.50000	1.75000	1.62500	0.66602
4	1.50000	1.62500	1.56250	0.25220
5	1.50000	1.56250	1.53125	0.05911

3.6.2 Secant Method

For the secant method, the initial approximations are chosen as

$$x_0 = 1, \qquad x_1 = 2.$$

The iteration formula is

$$x_{k+1} = x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}.$$

Table 3.2: Secant method for $f(x) = x^3 - x - 2$

Iteration	x_k	$f(x_k)$
0	1.00000	-2.00000
1	2.00000	4.00000
2	1.33333	-0.96296
3	1.46269	-0.33334
4	1.53117	0.05863
5	1.52093	-0.00269

Compared to bisection, the secant method moves toward the root more quickly. In this example, the iterates approach the solution efficiently without requiring the derivative of the function.

3.6.3 Newton's Method

For Newton's method, the initial guess is taken as

$$x_0 = 1.5.$$

Since

$$f(x) = x^3 - x - 2, \qquad f'(x) = 3x^2 - 1,$$

the iteration formula becomes

$$x_{k+1} = x_k - \frac{x_k^3 - x_k - 2}{3x_k^2 - 1}.$$

The convergence is very rapid. After only a few iterations, the approximation is already very close to the root. This behavior is consistent with the quadratic convergence of Newton's method near a simple root.

Table 3.3: Newton's method for $f(x) = x^3 - x - 2$

Iteration	x_k	$f(x_k)$
0	1.50000	-0.12500
1	1.52174	0.00214
2	1.52138	0.00000
3	1.52138	0.00000
4	1.52138	0.00000

3.6.4 Damped Newton Method

For the damped Newton method, the same function is used, but the Newton step is multiplied by a damping parameter. Here, the initial guess is chosen as

$$x_0 = 2,$$

and a constant damping parameter

$$\alpha = 0.8$$

is used for illustration. The iteration formula is

$$x_{k+1} = x_k - \alpha \frac{f(x_k)}{f'(x_k)}.$$

Table 3.4: Damped Newton method for $f(x) = x^3 - x - 2$

Iteration	x_k	$f(x_k)$
0	2.00000	4.00000
1	1.70909	1.28315
2	1.57686	0.34397
3	1.53426	0.07730
4	1.52406	0.01594
5	1.52192	0.00321

The method converges more cautiously than the classical Newton method. Although it is slower in this example, it provides a more controlled sequence of iterates and may be more reliable when the initial guess is not sufficiently good.

3.6.5 Brent's Method

Brent's method starts with a bracketing interval, so the initial interval $[1, 2]$ is again suitable. The method combines bisection, secant, and inverse quadratic interpolation steps. Because of this hybrid structure, it usually converges faster than bisection while preserving the safety of a bracketing method.

A possible sequence of approximations for this example is shown below.

The method reaches the root in a small number of iterations. In this example, it is both faster than bisection and more robust than the open methods.

Table 3.5: Brent's method for $f(x) = x^3 - x - 2$

Iteration	Approximation	$f(x_k)$
1	1.33333	-0.96298
2	1.58280	0.38252
3	1.51188	-0.05605
4	1.52094	-0.00261
5	1.52138	0.00000

3.6.6 Comparison for the Example

The numerical results obtained above show clear differences among the methods. Bisection is the slowest method, but it behaves in a very predictable way. Secant converges faster and avoids derivative evaluation. Newton's method gives the fastest convergence in this example, while damped Newton sacrifices some speed in exchange for additional stability. Brent's method combines reliability and efficiency in a balanced manner.

Table 3.6: General comparison of the methods for the illustrative example

Method	Initial Value(s)	Approximate Root	General Behavior
Bisection	$[1, 2]$	1.52138	Reliable but slow
Secant	$x_0 = 1,\ x_1 = 2$	1.52138	Faster, derivative-free
Newton	$x_0 = 1.5$	1.52138	Very fast near the root
Damped Newton	$x_0 = 2,\ \alpha = 0.8$	1.52138	Safer, but slower
Brent	$[1, 2]$	1.52138	Robust and efficient

3.6.7 Failure Cases

Bisection Method. The bisection method requires an initial interval where the function changes sign. Consider

$$f(x) = x^2 + 1.$$

For any real numbers a and b, we have $f(a) > 0$ and $f(b) > 0$. For example, on the interval $[-1, 1]$,

$$f(-1) = 2, \qquad f(1) = 2.$$

Since there is no sign change, the method cannot be applied and no iterations can be performed.

Secant Method. The secant method may fail when two consecutive function values are very close, leading to a nearly zero denominator. Consider

$$f(x) = (x - 1)^2,$$

with initial guesses

$$x_0 = 0.9, \qquad x_1 = 1.1.$$

Then

$$f(x_0) = 0.01, \qquad f(x_1) = 0.01.$$

The update formula

$$x_{k+1} = x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}$$

contains the denominator $f(x_k) - f(x_{k-1})$, which becomes zero in this case. As a result, the method breaks down and cannot proceed.

Newton's Method. Newton's method may diverge when the derivative is small. Consider

$$f(x) = x^3,$$

with initial guess $x_0 = 0.1$. Then

$$f'(x) = 3x^2, \qquad f'(0.1) = 0.03.$$

The first iteration gives

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.1 - \frac{0.001}{0.03} \approx 0.0667.$$

Subsequent iterations produce only small changes:

$$x_2 \approx 0.0444, \quad x_3 \approx 0.0296.$$

The convergence becomes very slow because the derivative is close to zero near the root.

Damped Newton Method. The damped Newton method may become inefficient if the step size is reduced too aggressively. Consider again

$$f(x) = x^3 - x - 2,$$

with initial guess $x_0 = 2$ and a small damping parameter, for example $\alpha = 0.1$.

The Newton step at x_0 is approximately -0.3636, so the damped update becomes

$$x_1 = 2 + 0.1 \cdot (-0.3636) \approx 1.9636.$$

The reduction in the function value is limited:

$$f(2) = 4, \qquad f(1.9636) \approx 3.600.$$

Compared to the standard Newton method, which moves directly close to the root, the damped version progresses much more slowly. Repeated small steps lead to inefficient convergence.

Brent's Method. Brent's method requires a valid bracketing interval. Consider again

$$f(x) = x^2 + 1.$$

On the interval $[-1, 1]$,

$$f(-1) = 2, \qquad f(1) = 2,$$

so there is no sign change. Since Brent's method relies on bracketing, it cannot be initialized in this case. Even though the method is robust, it still depends on the existence of an initial interval containing a root.

3.7 Comparison of Methods

In this section, the numerical methods discussed for solving single nonlinear equations are compared in terms of convergence speed, robustness, and computational requirements.

The bisection method is the most reliable among all the methods considered. As long as the initial interval satisfies the sign change condition and the function is continuous, convergence to a root is guaranteed. However, its linear rate of convergence makes it relatively slow, especially when high accuracy is required [1, 2].

The secant method improves upon the bisection method by achieving a faster, superlinear convergence rate without requiring the computation of derivatives. This makes it more efficient in many cases. However, it does not guarantee convergence and may fail if the initial guesses are not chosen appropriately [3].

Newton's method offers the fastest convergence among the methods discussed, with quadratic convergence under suitable conditions. When the initial guess is sufficiently close to the root and the derivative is well-behaved, the method is highly efficient. Nevertheless, its performance is sensitive to the initial guess, and it requires the evaluation of the derivative, which may not always be practical [1, 3].

The damped Newton method improves the robustness of the classical Newton method by introducing a step size parameter. This modification helps prevent divergence and allows the method to converge from a wider range of initial guesses. However, this increased stability often comes at the cost of slower convergence compared to the standard Newton method [3].

Brent's method combines the strengths of both bracketing and open methods. It maintains the guaranteed convergence of the bisection method while incorporating faster interpolation steps similar to the secant method. As a result, it provides a good balance between robustness and efficiency and is often preferred in practical applications [5, 4].

Overall, there is a trade-off between convergence speed and robustness. Methods such as Newton's method are very fast but sensitive to initial conditions, while methods like bisection are slower but more reliable. Hybrid approaches such as Brent's method aim to achieve both reliability and efficiency, making them particularly useful in real-world problems.

4. Systems of Nonlinear Equations

4.1 Multivariable Newton Method

Newton's method can be extended to systems of nonlinear equations in a natural way. Consider a system of the form

$$F(\mathbf{x}) = \mathbf{0}, \tag{4.1}$$

where $F : \mathbb{R}^n \to \mathbb{R}^n$ is a vector-valued function.

Starting from an initial guess $\mathbf{x}_0$, the method generates a sequence $\{\mathbf{x}_k\}$ using the iterative formula

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{s}_k, \tag{4.2}$$

where the step $\mathbf{s}_k$ is obtained by solving the linear system

$$J_F(\mathbf{x}_k)\mathbf{s}_k = -F(\mathbf{x}_k). \tag{4.3}$$

Here, $J_F(\mathbf{x}_k)$ denotes the Jacobian matrix evaluated at $\mathbf{x}_k$. This formulation can be interpreted as a multivariable generalization of the Newton update, where the nonlinear system is locally approximated by a linear one.

At each iteration, a linear system must be solved. Therefore, compared to the scalar case, the computational cost per iteration is higher. However, under suitable conditions, the method retains its rapid convergence properties.

If the initial guess is sufficiently close to the solution and the Jacobian matrix is nonsingular at the root, the method typically exhibits quadratic convergence. This makes it one of the most efficient methods for solving nonlinear systems [3, 1].

Despite its strong convergence properties, the method has several limitations. The need to compute and factorize the Jacobian matrix can be computationally expensive, especially for large systems. In addition, if the Jacobian is singular or nearly singular, the method may fail or produce unstable steps.

For these reasons, modifications and alternative methods are often considered in practice, particularly when dealing with large-scale or ill-conditioned systems.

4.2 Convergence Analysis

The convergence behavior of Newton's method for systems is similar to the scalar case, but it depends on additional factors related to the structure of the system.

Let $\mathbf{x}^*$ be a solution of the system $F(\mathbf{x}) = \mathbf{0}$. If the function F is sufficiently smooth and the Jacobian matrix $J_F(\mathbf{x}^*)$ is nonsingular, then Newton's method exhibits quadratic convergence in a neighborhood of $\mathbf{x}^*$. This means that the error satisfies

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\| \approx C\|\mathbf{x}_k - \mathbf{x}^*\|^2, \tag{4.4}$$

for some constant $C > 0$ and sufficiently large k [3].

This rapid convergence makes Newton's method very efficient when a good initial approximation is available. However, the method is not globally convergent. If the initial guess is far from the solution, the method may fail to converge or may converge to an unintended solution.

Another important factor is the conditioning of the Jacobian matrix. If $J_F(\mathbf{x}_k)$ is ill-conditioned or nearly singular, solving the linear system

$$J_F(\mathbf{x}_k)\mathbf{s}_k = -F(\mathbf{x}_k) \tag{4.5}$$

may lead to large or unstable steps. This can cause divergence or oscillatory behavior.

In practice, the convergence of the method is often monitored using a combination of criteria such as

$$\|\mathbf{F}(\mathbf{x}_k)\| < \varepsilon \tag{4.6}$$

and

$$\|\mathbf{x}_{k+1} - \mathbf{x}_k\| < \varepsilon, \tag{4.7}$$

where ε is a prescribed tolerance.

Due to these limitations, various modifications of Newton's method have been developed to improve its robustness. These include damping strategies, line search techniques, and quasi-Newton methods such as Broyden's method.

4.3 Broyden's Method

Broyden's method is a quasi-Newton method developed for solving systems of nonlinear equations. The main idea of the method is to avoid the explicit computation of the Jacobian matrix while still retaining a structure similar to Newton's method.

In Newton's method, the step $\mathbf{s}_k$ is obtained by solving

$$J_F(\mathbf{x}_k)\mathbf{s}_k = -F(\mathbf{x}_k). \tag{4.8}$$

In contrast, Broyden's method replaces the Jacobian matrix with an approximation B_k, leading to the system

$$B_k\mathbf{s}_k = -F(\mathbf{x}_k). \tag{4.9}$$

The approximation B_k is updated at each iteration using information from previous steps. One common update formula, known as the good Broyden update, is given by

$$B_{k+1} = B_k + \frac{(\mathbf{y}_k - B_k\mathbf{s}_k)\mathbf{s}_k^T}{\mathbf{s}_k^T\mathbf{s}_k}, \tag{4.10}$$

where

$$\mathbf{y}_k = F(\mathbf{x}_{k+1}) - F(\mathbf{x}_k). \tag{4.11}$$

This update ensures that the new approximation B_{k+1} satisfies a secant-like condition, which is a generalization of the secant method to higher dimensions.

One of the main advantages of Broyden's method is that it eliminates the need to compute the Jacobian matrix explicitly. This can significantly reduce computational cost, especially for large systems or when evaluating derivatives is expensive.

However, the convergence of Broyden's method is generally slower than that of Newton's method. While Newton's method achieves quadratic convergence under suitable conditions, Broyden's method typically exhibits superlinear convergence.

The performance of the method also depends on the choice of the initial approximation B_0. A good initial approximation can improve convergence speed, while a poor choice may lead to slow convergence or instability.

Despite these limitations, Broyden's method is widely used in practice due to its efficiency and its ability to handle problems where the Jacobian is difficult to compute [3].

4.4 Illustrative Example

To illustrate the behavior of the methods for systems of nonlinear equations, consider the following system:

$$\begin{cases} x^2 + y^2 - 4 = 0, \\ x - y - 1 = 0. \end{cases} \tag{4.12}$$

This system represents the intersection of a circle and a line. One of its solutions is approximately

$$\mathbf{x}^* \approx (1.82288,\ 0.82288).$$

The same system is used to demonstrate both Newton's method and Broyden's method. The goal is to observe how the iterates approach the solution.

4.4.1 Newton Method for Systems

For Newton's method, the Jacobian matrix is given by

$$J_F(\mathbf{x}) = \begin{bmatrix} 2x & 2y \\ 1 & -1 \end{bmatrix}. \tag{4.13}$$

Starting from the initial guess

$$\mathbf{x}_0 = (2,\ 1),$$

the method generates a sequence of approximations.

Table 4.1: Newton method for the system

Iteration	x_k	y_k	$\|F(\mathbf{x}_k)\|$
0	2.00000	1.00000	1.00000
1	1.83333	0.83333	0.05556
2	1.82292	0.82292	0.00022
3	1.82288	0.82288	0.00000

The method converges rapidly to the solution. After only a few iterations, the residual becomes very small. This behavior reflects the quadratic convergence of Newton's method near the root.

4.4.2 Broyden's Method for Systems

For Broyden's method, the same initial guess is used:

$$\mathbf{x}_0 = (2,\ 1).$$

The initial Jacobian approximation is taken as the identity matrix.

4.4.3 Broyden's Method for Systems

For Broyden's method, the same initial guess is used:

$$\mathbf{x}_0 = (2,\ 1).$$

In order to obtain a stable and informative sequence, the initial Jacobian approximation is chosen as

$$B_0 = \begin{bmatrix} 4 & 2 \\ 1 & -1 \end{bmatrix},$$

which corresponds to the exact Jacobian at the initial point.

Table 4.2: Broyden method for the system

Iteration	x_k	y_k	$\|F(\mathbf{x}_k)\|$
0	2.00000	1.00000	1.00000
1	1.83333	0.83333	0.05556
2	1.82353	0.82353	0.00346
3	1.82288	0.82288	0.00001
4	1.82288	0.82288	0.00000

The method converges more slowly than Newton's method, but it still reaches the solution efficiently. In this example, the use of an initial Jacobian approximation close to the true Jacobian improves the stability of the iterates and leads to a smooth convergence pattern.

4.4.4 Comparison for the Example

The results show that Newton's method converges faster than Broyden's method in this example. This is expected, since Newton's method uses exact derivative information, while Broyden's method relies on approximations.

On the other hand, Broyden's method avoids the computation of the Jacobian matrix, which can be advantageous in problems where derivative evaluation is expensive.

4.4.5 Failure Cases

Newton Method. Newton's method may fail when the Jacobian matrix becomes singular or nearly singular. Consider the system

$$\begin{cases} x^2 = 0, \\ y = 0. \end{cases} \tag{4.14}$$

The Jacobian matrix is

$$J_F(x, y) = \begin{bmatrix} 2x & 0 \\ 0 & 1 \end{bmatrix}.$$

At the point $(0, y)$, the Jacobian is singular. Starting from

$$\mathbf{x}_0 = (0, 2),$$

we have

$$F(\mathbf{x}_0) = (0, 2), \qquad J_F(\mathbf{x}_0) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

The Newton step requires solving

$$J_F(\mathbf{x}_0)\mathbf{s}_0 = -F(\mathbf{x}_0),$$

which becomes

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{s}_0 = \begin{bmatrix} 0 \\ -2 \end{bmatrix}.$$

This system has no unique solution, since the Jacobian matrix is singular. Therefore, the method cannot proceed from this initial point.

Even when the Jacobian is not exactly singular but close to singular, the computed step may become very large, leading to instability and divergence.

Broyden Method. Broyden's method may fail when the update of the Jacobian approximation becomes numerically unstable. Consider again the system

$$\begin{cases} x^2 + y^2 - 4 = 0, \\ x - y - 1 = 0, \end{cases} \tag{4.15}$$

and assume that two consecutive iterates are very close:

$$\mathbf{x}_k = (1.82288,\ 0.82288), \qquad \mathbf{x}_{k+1} = (1.82289,\ 0.82289).$$

Then the step

$$\mathbf{s}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$$

is extremely small. In the Broyden update formula

$$B_{k+1} = B_k + \frac{(\mathbf{y}_k - B_k\mathbf{s}_k)\mathbf{s}_k^T}{\mathbf{s}_k^T\mathbf{s}_k},$$

the denominator $\mathbf{s}_k^T\mathbf{s}_k$ becomes very small.

As a result, the update term may become very large, causing a significant distortion in the Jacobian approximation. This can lead to unstable iterations and loss of convergence.

In practice, such situations are handled by introducing safeguards, such as skipping the update or restarting the method with a new approximation.

4.5 Comparison of Methods

In the context of systems of nonlinear equations, Newton's method and Broyden's method exhibit different trade-offs.

Newton's method is generally more accurate and converges faster when the Jacobian matrix is well-behaved and the initial guess is sufficiently close to the solution. However, the need to compute and solve systems involving the Jacobian matrix increases the computational cost.

Broyden's method, on the other hand, reduces computational effort by avoiding explicit Jacobian evaluations. This makes it more suitable for large-scale problems or problems where derivative information is difficult to obtain.

Overall, Newton's method is preferred when accuracy and fast convergence are critical, while Broyden's method is advantageous when computational efficiency is more important [3, 5].

Plans for the Next Term

In the next term, the focus will be on extending the study to systems of nonlinear equations, particularly through methods such as multivariable Newton and Broyden methods. The necessary theoretical background for these methods will be studied in detail, and their practical implementation will be carried out using a computational tool.

Furthermore, numerical experiments will be performed to compare the selected methods in terms of convergence behavior, computational efficiency, and robustness. Finally, the relationship between root-finding methods and optimization techniques will be explored to provide additional insight, and the thesis will be completed in accordance with academic standards.

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